

PHASE 1 — FOUNDATION

Week 5 — Control Box I

PufferFish first. Then we touch the real board.

Say Watt Robotics · ROV Team | 2026-2027 MATE ROV Season

MEETING 5 · THREE HOURS · MADISON, ALABAMA

AGENDA

Three hours. Practice board, then the real board.

0:00	Welcome & Safety Brief	15 min
0:15	The PufferFish — Practice Board Brief	15 min
0:30	PufferFish — Hands On	45 min
1:15	The TriggerFish REV 4 — Where We Pick Up	15 min
1:30	Break	10 min
1:40	TriggerFish — Next Block of Components	50 min
2:30	Meter-Check Discipline	15 min
2:45	Wrap-Up & Homework	15 min

We run on time. Every block has a clock, and we hold to it — that habit is part of the engineering.

TODAY'S ROLES

Four jobs. Four students. Different every week.

Every meeting we assign these four — by season's end, every student has done every job.

NOTETAKER / SCRIBE

TODAY: _____

Captures every decision, action item, and blocker. Files notes with the CFO for the engineering notebook.

SAFETY BRIEFER

TODAY: _____

Delivers the standing four-question safety brief at the start of the meeting.

PHOTOGRAPHER

TODAY: _____

Captures photos and short videos throughout — feeds the engineering notebook and the marketing display.

TIMEKEEPER

TODAY: _____

Holds the meeting to the agenda's clock. Calls the time at each block change.

SECTION 01

Welcome & Safety Brief

0:00 - 0:15 · 15 MIN

Same ritual every meeting. We get sharper at it; the kids start running it.



Every meeting opens with a safety brief.

Why we do this

Real engineering and dive teams brief safety before every job — not because they expect to fail, but because the discipline of asking the same four questions every time is what makes a team safe in the long run. We start the habit now, when the stakes are small, so it's automatic when they aren't.

- 1 **What are we doing today?**
- 2 **What could hurt someone?**
- 3 **How do we prevent it?**
- 4 **Who do we tell if something happens?**

TODAY'S BRIEF · MEETING 5

THE WORK

Soldering through-hole components onto two boards: the SeaMATE PufferFish practice board and the TriggerFish REV 4 control box.

THE HAZARDS

Iron tip at 650°F. Back of the board and component leads are hot too. Metal clippings fly when cutting leads. Lead-based solder.

PREVENTION

Iron in holder when not in use. Fume extractor on, fan blowing away from the student. Safety glasses for snipping. Wash hands when done.

IF SOMETHING HAPPENS

Stop. Speak up. Tell the advisor immediately. Burns get cool water for 15 minutes. Loose iron on the bench? Call it out.

LOOKING AHEAD

1.6 is wiring and power — fewer joints, bigger consequences if a polarity check gets skipped. Habits we set today carry forward.

SECTION 02

Practice First

0:15 - 1:15 · 60 MIN

The PufferFish board is a low-stakes circuit. We earn the technique here before we touch the real one.



PUFFERFISH

Earn a clean joint first.

The SeaMATE PufferFish board has nothing to do with the ROV. That's why we use it.

The PufferFish is a small practice circuit MATE built for one purpose: give every kid a low-stakes board to develop the technique on before they sit down at the TriggerFish.

If you mess up a PufferFish joint, you re-flow it, re-do it, or pull the component and try again. If you mess up a TriggerFish joint — a \$290 component on a board you only get one of — it's a different conversation.

The frame for the night: working vs. crafted. A working joint passes a multimeter check. A crafted joint also survives heat, vibration, and the trip to the pool. The PufferFish is where we calibrate what crafted feels like.

Every joint is the same five moves.

1

Seat the component flush

Bend leads, push through, flip the board, make sure the body is flat against the PCB. A component sitting half a millimeter off is a future failure.

2

Heat the joint, not the solder

Iron tip on the pad AND the lead at the same time, for one full second. The pad has to be hot enough to flow solder on its own.

3

Feed solder into the joint

Touch the solder to the joint — not the iron tip. If it doesn't flow, the joint isn't hot enough yet.

4

Pull the solder, then the iron

Solder first, iron second. The order matters — pulling the iron first leaves a peak; pulling solder first leaves a clean dome.

5

Inspect

Shiny, smooth, concave. Dull or grainy means a cold joint. Re-flow it now, not later.

SECTION 03

TriggerFish

Where We Pick Up

1:15 - 1:30 · 15 MIN

1.3 left us with the fuse holder, the two 1000 μ F caps, and the 3 A fuse mounted. Here's the next block.



Where we left off — and what comes next.

From 1.3 the board already carries: the fuse holder, two 1000 μ F / 16V electrolytic capacitors (with polarity verified before commit), the 3 A fuse seated in the holder, and the power leads run through heatshrink.

Tonight we open the SeaMATE TriggerFish ROV Guide on the iPad and pick up at the next block of components per the slide sequence. The Guide is the team's reference of record — every step gets verified against the slide before any iron touches the board.

Polarity rule still applies: every electrolytic cap, every diode, every transistor — verify the orientation against the board silkscreen AND the Guide image before you commit the joint. A reversed electrolytic cap fails open, fails loud, and ends the night.

Two things on every slide.

THE INSTRUCTION

What the slide says to do. Read it out loud as a team before anyone picks up an iron. If the verb is 'solder,' name what gets soldered and in what order.

THE WARNING

What the slide flags. Polarity. Orientation. Sequence. Any line that says 'double check' or 'verify' gets a second pair of eyes before any commit.

BREAK

Ten minutes.

We resume in ten — and, like always, we start on time.

SECTION 04

The Next Block

1:40 - 2:30 • 50 MIN

Fifty minutes, hands on. Coach John runs the brief; the kids run the build.



How we work through the Guide.

1

Read the slide as a team

One person reads it out; the others listen. No iron picked up until the team agrees on what the slide is asking for.

2

Stage the components

Lay them out next to the board, oriented the way they'll go in. Polarity-sensitive parts marked clearly.

3

Solder, one student at a time

Whoever has the iron, owns that joint. Others coach, photograph, or cue up the next component.

4

Inspect and meter-check

Shiny, smooth, concave joint. Continuity to the next pad in the circuit; no shorts to ground or neighboring pads. If the meter doesn't agree, re-flow.

5

Move to the next slide

Only when the current step passes inspection AND meter-check. We don't get ahead of the Guide.

SECTION 05

Meter-Check Discipline

2:30 - 2:45 · 15 MIN

Three reads, every stage. Continuity good, no shorts, polarity confirmed.



THREE READS

Every completed stage gets metered before we move on.

The watt meter from 1.3 stays on the bench. The multimeter is the workhorse for these reads.

- 1 Continuity from each new joint to its intended next pad**
Beep means the joint connects where the schematic says it does. Silence means re-flow or re-route.
- 2 No shorts to ground or to neighboring pads**
Probe each new pad against ground and against the pads immediately adjacent. Beep here means a solder bridge — fix it before power.
- 3 Polarity confirmed against the Guide image**
Eyes on the silkscreen, eyes on the Guide, eyes on the component. Three references; one orientation. If two agree and one doesn't, stop and figure out which is wrong.

SECTION 06

Wrap-Up & Homework

2:45 - 3:00 · 15 MIN

Capture what we decided. Close the loop. Hand off the next move.



Before we meet again.

EVERY STUDENT

- 1 Re-read the SeaMATE TriggerFish ROV Guide slides we covered tonight. If you led a joint, be able to explain why the polarity went the way it did.
- 2 Bring closed-toe shoes again next week — we power the board up for the first time in 1.6.
- 3 If you didn't get a turn at the iron tonight, message the CEO so you're at the top of the rotation in 1.6.

THE PARALLEL TRACK · LEADS & ADVISORS

Advisors: confirm the 12V SLA battery, charger, and power leads are in hand by Wednesday — they're the gating items for 1.6's smoke test. Confirm the pre-terminated tether arrived; if not, escalate the order.

NEXT MEETING · WEEK 6

Control Box II

Finish the wiring. Build the ROV's power wire. Connect the tether. Fuse the circuit to measured full-load amps. Meter-check every stage before first power-up.

ACTION ITEMS

What we decided. Who owns it. When it's due.

The Notetaker fills these in during the meeting and reads them back before we leave.

ITEM	OWNER	DUE

Tools accounted for. Shop reset.

We do this together, every meeting. No exceptions — the Tool Steward leads.

- ☐ Soldering irons unplugged, cool, and put away.
- ☐ PufferFish boards collected; loose components back in the kit.
- ☐ TriggerFish board secured in its tray; iPad with the Guide returned.
- ☐ Multimeter and watt meter back in their cases.
- ☐ Workshop swept; benches wiped.
- ☐ Today's photos uploaded to the team folder.
- ☐ Notetaker's meeting notes filed with the CFO for the engineering notebook.

PRACTICE EARNED. REAL WORK STARTED.

Next week, we power it up.

The PufferFish taught us the technique. The TriggerFish is where we use it. 1.6 closes the loop with the first smoke test.